

Advanced School Management SaaS Platform - Complete Module Blueprint

This document provides a complete enterprise-level School Management SaaS architecture with all major and advanced modules required for modern schools, colleges, coaching institutes, universities, and multi-campus educational organizations.

1. Super Admin SaaS Panel

- Multi-school management
 - Subscription & plan management
 - School onboarding
 - Usage analytics
 - Global role & permissions
 - Feature toggles
 - Custom branding/white-label
 - Billing & invoices
 - Tax/GST support
 - API management
 - Server monitoring
 - Activity logs
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2. School Admin Module

- School profile setup
 - Academic year management
 - Department management
 - Campus/branch management
 - Class & section management
 - Staff management
 - Academic settings
 - Document settings
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3. Student Management

- Online admission

- Student onboarding
 - Student profile
 - Guardian information
 - Student ID cards
 - Digital documents
 - Transfer certificate
 - Migration records
 - Disciplinary records
 - Medical records
 - Student lifecycle tracking
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4. Attendance System

- RFID attendance
 - Biometric attendance
 - Face recognition attendance
 - QR attendance
 - Geo-fencing attendance
 - Teacher attendance
 - Bus attendance
 - Real-time notifications
 - Attendance analytics
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5. Learning Management System (LMS)

- Course management
 - Video lectures
 - Recorded sessions
 - Live classes integration
 - Assignments
 - Quizzes
 - Homework tracking
 - Discussion forums
 - Notes & resources
 - AI-powered learning recommendations
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6. Examination & Result System

- Exam scheduling
 - Online exams
 - Question banks
 - OMR support
 - Automatic grading
 - Result generation
 - Rank generation
 - Progress analytics
 - Report cards
 - AI-based performance insights
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7. Fee & Finance Management

- Fee structures
 - Online payment gateway
 - Installment support
 - Scholarships
 - Late fee automation
 - GST invoices
 - Accounting integration
 - Payroll
 - Expense tracking
 - Financial dashboards
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8. Parent Portal

- Real-time student tracking
 - Attendance reports
 - Fee payments
 - Homework updates
 - Exam results
 - Communication with teachers
 - School notices
 - Transport tracking
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9. Teacher Portal

- Class timetable
 - Attendance marking
 - Assignment creation
 - Exam evaluation
 - Resource uploads
 - Communication tools
 - Performance reports
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10. AI & Automation Features

- AI chatbot for parents/students
 - Automated timetable generation
 - Smart attendance anomaly detection
 - Performance prediction
 - AI report summaries
 - Voice assistant
 - AI-powered doubt solving
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11. Transport Management

- Bus route management
 - GPS live tracking
 - Driver management
 - Pickup/drop notifications
 - Transport fee management
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12. Hostel Management

- Room allocation
 - Hostel attendance
 - Mess management
 - Visitor tracking
 - Hostel fee management
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13. Library Management

- Book catalog

- Barcode scanning
 - Issue/return tracking
 - Fine calculation
 - Digital library
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14. Communication Module

- Push notifications
 - SMS integration
 - Email notifications
 - WhatsApp integration
 - Emergency alerts
 - Announcement system
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15. HR & Payroll

- Employee onboarding
 - Leave management
 - Payroll generation
 - Salary slips
 - Tax deductions
 - Performance reviews
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16. Security & Compliance

- Role-based access control
 - 2FA authentication
 - Audit logs
 - Data encryption
 - GDPR compliance
 - Backup systems
 - Disaster recovery
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17. Mobile Applications

- Student app
- Parent app

- Teacher app
 - Admin app
 - Offline support
 - Push notifications
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18. Analytics & Reports

- Student performance analytics
 - Attendance trends
 - Financial reports
 - Custom report builder
 - Predictive analytics
 - Executive dashboards
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19. Advanced SaaS Features

- Multi-tenant architecture
 - Subdomain support
 - White-label SaaS
 - Marketplace integrations
 - API-first architecture
 - Usage-based billing
 - Custom workflows
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20. Recommended Tech Stack

- Frontend: React / Next.js
- Mobile: React Native
- Backend: Node.js / NestJS / FastAPI
- Database: PostgreSQL + Redis
- Storage: AWS S3
- Real-time: Socket.IO / WebSocket
- Queue: Kafka / RabbitMQ
- Containerization: Docker
- Cloud: AWS / Azure / GCP

Suggested Enterprise Architecture

Recommended architecture for a scalable school SaaS platform:

Layer	Recommended Technology
Frontend	Next.js + Tailwind CSS
Mobile	React Native
API Gateway	NGINX + API Gateway
Backend	NestJS / FastAPI Microservices
Database	PostgreSQL
Caching	Redis
Queue	Kafka / RabbitMQ
File Storage	AWS S3
Authentication	JWT + OAuth + 2FA
CI/CD	GitHub Actions + Docker
Monitoring	Prometheus + Grafana

Development Roadmap

- Phase 1: Authentication, SaaS onboarding, student & teacher modules
- Phase 2: Attendance, fees, exams, parent portal
- Phase 3: LMS, transport, hostel, library
- Phase 4: AI automation, analytics, reporting
- Phase 5: Mobile apps, scaling, white-label support

This blueprint is suitable for building a modern enterprise-grade SaaS School Management System comparable to global education ERP platforms.